

Signal Harbour

Systematic multi-asset funds, a sector news-sentiment index, and a Streamlit research app

FINS3645 Project B

Kaiyuan Lan (z5444541)

Sample through December 2023

1. Customer, data, and backtest design

Signal Harbour is designed for the portfolio analyst in a small wealth-advisory firm who has to turn model output into an investment-committee recommendation. The practical gap is not a shortage of optimisation methods, it is the lack of one place where equity and crypto proposals can be compared on the same wealth, drawdown and holdings views, and then explained to a client. The product therefore presents Equity, Crypto and Combined sleeves for equal weight, minimum variance, maximum Sharpe and risk parity. Markowitz (1952) provides the portfolio framework; the news work starts from the VADER model of Hutto and Gilbert (2014).

Calendar handling is settled before the funds are built. Simple returns are calculated from adjusted close inside each ticker, leaving weekend crypto observations on their native dates. Equity and Combined funds use 252 observations for estimation and annualisation; stand-alone Crypto funds use 365. For a Combined fund, the already calculated crypto return is then left-aligned to the equity calendar. Aligning prices first would roll several weekend moves into Monday and misdescribe that compound move as a one-day return. The adopted sequence avoids this problem, although it means that the Combined sleeve is evaluated only when equities trade.

The monthly decision rule follows the same timing discipline. At date T , the optimiser receives data only through $T-1$; the existing holdings earn return T , and the new targets take effect at the next session. Holdings then drift with relative asset performance until the following decision. All funds are long-only, and the broad sleeves impose a 25% cap on each name. Covariance is annualised inside the solver, but numerical convergence alone is not accepted: the candidate must improve on equal weight for the stated objective.

Backtest conventions

A zero risk-free rate is used in both the Maximum Sharpe objective and the reported Sharpe ratios. Table 1 shows fund performance before trading costs, whereas Table 5 subjects the Minimum Variance base and its sentiment overlay to identical turnover charges. The tables consequently answer different questions. Estimation is also kept out of the Streamlit interface: the app reads retained CSV outputs instead of rerunning VADER, rebuilding sentiment or repeating the backtests on page load.

First-live timing

After the 252-session warm-up, Equity and Combined weights are first selected on 4 January 2021 and begin earning on 5 January. The corresponding native-crypto dates are 1 and 2 January. Figure A1 starts at those first earnable returns, so the 2020 estimation history is not presented as if a portfolio had already been live.

Why timing and drift are specified this way

A portfolio formed after the close cannot claim the close-to-close return that has just been observed. Leaving return T with the old holdings and switching at the next session removes that extra day of information. The dated weight file makes the convention directly auditable.

Drift addresses a separate issue. Resetting the book to target weights every day would quietly convert a monthly strategy into a daily one, even if the optimiser ran only monthly; turnover is therefore measured from the holdings after market movement. The solver required an additional check. When daily covariance was left unscaled, SLSQP could announce convergence without moving far from its starting vector. Annualising the objective and comparing the solution with equal weight revealed the stall. In the retained Equity sample, Minimum Variance differs from Equal Weight at every decision date, and their mean half-L1 distance is about 0.72. That is a material change in the portfolio, rather than a rounding artefact.

2. Out-of-sample fund results

Table 1 does not identify a single winner on every measure. Within the mixed-asset group, Combined Risk Parity has the strongest risk-adjusted record: a Sharpe ratio of 0.91, 14.3% CAGR, 16.2% volatility and a -19.3% maximum drawdown (Figures A2 and A4). Combined Maximum Sharpe finishes marginally higher, at 1.55 times initial wealth and a 15.7% CAGR, although its drawdown reaches -26.2%. An adviser would accept a noticeably larger loss and more concentrated holdings for a small increase in terminal growth.

How to read the mixed-asset ranking

Diversification is only part of the explanation because the trading calendar changes the return sequence. A stand-alone crypto fund compounds across all 365 days. In a Combined fund, the crypto return is calculated on that native calendar but enters the portfolio only on equity sessions. Risk Parity then reduces the weight of assets that appear most volatile in the trailing covariance matrix. Since it does not estimate expected returns, it avoids one especially fragile input in a live sample of only three years.

Maximum Sharpe must estimate both mean returns and covariance. With three live years, small changes in the estimated means can translate into large positions. Table 3 makes that concentration visible at December 2023, while Figure A2 records the deeper drawdown along the way. Combined Risk Parity is consequently the more defensible opening point for a client discussion. Maximum Sharpe remains relevant as a satellite option for a client who is prepared to tolerate a more concentrated book and a larger temporary loss.

Equity and crypto sleeves

The equity-only results are less favourable to optimisation. Equal Weight leads with a 0.86 Sharpe ratio, ahead of Risk Parity at 0.76 and Minimum Variance at 0.47. In this walk-forward period, the additional estimation did not outperform the transparent baseline. Minimum Variance does not forecast means, so its weaker result is more plausibly associated with the trailing covariance estimate and binding name caps than with the mean-estimation error that affects Maximum Sharpe.

Crypto Minimum Variance is the strongest crypto sleeve, with a 1.04 Sharpe ratio and 62.1% CAGR. Crypto Maximum Sharpe illustrates why CAGR and Sharpe must be read together: its annualised arithmetic mean-to-volatility ratio is positive at 0.35, yet compounded CAGR is -2.7%. Several unusually strong days can raise the arithmetic mean even though volatility and the order of returns leave final wealth below its starting value.

Table 1. Out-of-sample performance. CAGR is the annualised return from the growth-of-one path. Sharpe is annualised arithmetic mean divided by annualised volatility, risk-free rate zero. Equity-calendar funds use 252 days; native crypto funds use 365. Source: performance_metrics.csv. See Figures A1 and A4.

Fund	CAGR	Vol.	Sharpe	MaxDD	Terminal
Eq EW	13.3%	16.1%	0.86	-20.2%	1.45x
Eq MinVar	5.4%	12.7%	0.47	-14.9%	1.17x
Eq MaxSharpe	7.0%	17.5%	0.48	-22.8%	1.22x
Eq RP	10.4%	14.5%	0.76	-18.4%	1.34x
Cry EW	37.1%	81.7%	0.80	-81.7%	2.58x
Cry MinVar	62.1%	71.2%	1.04	-71.5%	4.25x
Cry MaxSharpe	-2.7%	76.2%	0.35	-85.6%	0.92x
Cry RP	41.0%	79.7%	0.83	-80.1%	2.80x
Comb EW	15.5%	21.5%	0.78	-27.7%	1.54x
Comb MinVar	5.4%	12.8%	0.48	-15.1%	1.17x

Fund	CAGR	Vol.	Sharpe	MaxDD	Terminal
Comb MaxSharpe	15.7%	23.3%	0.74	-26.2%	1.55x
Comb RP	14.3%	16.2%	0.91	-19.3%	1.49x
Eq MinVar+Sent	5.3%	12.7%	0.47	-14.9%	1.17x

Concentration and the Combined sleeve

Figure A3 plots the decision weights of Combined Minimum Variance and Risk Parity, placing all unshown holdings in Other so the areas sum to one. Risk Parity remains relatively dispersed, consistent with the shallower drawdown in Figure A2. Minimum Variance clusters more heavily in names that appeared quiet in the trailing covariance matrix. This follows from its objective, but the apparent safety depends on the selected cluster continuing to behave as it did during estimation.

Across the menu, the Combined sleeve moderates native-crypto volatility without removing crypto exposure. Pure Crypto Equal Weight and Risk Parity end with much greater wealth, but their drawdowns are extreme; Crypto Maximum Sharpe also combines a positive Sharpe ratio with a negative CAGR. The alternatives therefore suit different clients. Combined Risk Parity is a plausible default on the equity-session calendar, whereas a pure-crypto fund would require a separate discussion with a substantially more risk-tolerant investor.

Table 2. Latest Combined Risk Parity decision weights, top five. Source: latest_holdings_snapshot.csv.

Ticker	Weight
MRK	3.81%
ABBV	3.74%
WMT	3.37%
KO	3.19%
TMUS	3.17%

Table 3. Latest Combined Maximum Sharpe decision weights, top five. Source: latest_holdings_snapshot.csv.

Ticker	Weight
GE	25.00%
NVDA	20.09%
SO	16.86%
ADBE	10.32%
BTC-USD	10.29%

These tables are end-of-sample snapshots. They describe concentration at the final decision date, not the full path from 2021 to 2023. A large December 2023 weight cannot explain performance earlier in the sample.

3. Sentiment model, sector index, and fusion

Part B retains the headline assembly established in Part A. Exact ticker-date-title duplicates are removed; each remaining headline is assigned to the current equity session, or the next session if the market is closed. Unmodified Base VADER and Week 9 finVADER scores provide the comparison points for the project extensions. The Week 9 analyser combines NLTK VADER, SentiBigNomics scaled by 0.1 and the Henry lexicon, matching the course helper and Korab's FinVADER resources. Its finance vocabulary also addresses the problem identified by Loughran and McDonald (2011): an apparently negative word in ordinary language may have a different meaning in financial text.

Signal Harbour extends the Week 9 score with documented masks, additional finance terms and negation-aware phrases. Coverage and the elevated-volume component of Part A's Attention Pulse then modify confidence in the score. Only unusual increases in news volume can raise confidence; quiet days do not receive a bonus. Missing ticker-days are restored as neutral before sector averaging, so Figure A5 and the portfolio signal share one no-news convention. A one-session lag is applied before the score becomes tradable.

The sector index as an economic series

Figure A5 tracks the lagged tone of the headline panel rather than a stand-alone market-timing rule. Sector averages are mildly positive over 2020-2023, but their ordering shifts. During the February-April 2020 COVID stress window, Energy is lowest around the figure of 0.009, while Technology and Real Estate are nearer 0.05. The pattern is compatible with the energy-demand shock and relatively resilient wording in the supplied Technology and Real Estate headlines. It remains a textual interpretation: the index neither measures realised sector returns nor identifies why the language changed.

The ordering changes by 2022. Energy rises to roughly 0.061, above Materials and Communications at around 0.023-0.030. Inflation, stronger energy cash flows and supply concerns offer an economically plausible reading of that shift, although they do not validate the index. Healthcare also moves from approximately 0.036 in 2021 to 0.053 in 2023, and Technology recovers from 0.033 in 2022 to 0.051 in 2023.

Figure A11 places these movements against uneven news coverage. Technology and Consumer average more than five headlines on news-bearing ticker-days; Materials and Real Estate are closer to two and have larger shares of exact-zero raw scores. Context weighting reduces the influence of sparse observations without making coverage equal. Communications remains within 0.027-0.038, Energy moves from 0.017 in 2020 to 0.061 in 2022 and back to 0.041 in 2023, and Consumer stays near 0.037-0.045. Because neutral filling draws every series towards zero, changes within a sector and comparisons across years are more informative than the absolute level of one line.

Continuity with Part A

Attention Pulse connects the Part B model with Part A, but only through volume confidence. Unusually high headline activity can strengthen a score; unusually low activity cannot. The result is still lagged before it reaches a portfolio. Since Part A did not establish Attention Pulse as a return signal, treating it as a same-day trigger would exceed the evidence.

The audit record separates the course resources from the project additions. Week 9 FinVADER keeps its Apache-2.0 notices, while the new terms, phrases and masks are identified as Signal Harbour components. That provenance is more useful than calling the model 'augmented finVADER', a label that does not explain why a headline's score changed.

Distributional model comparison

Table 4 and Figure A8 compare positive, neutral and negative shares across 4,000 headlines. Signal Harbour classifies fewer cases as neutral than Week 9 finVADER, but this distributional change does not establish better classification. The 120 worksheet was manually reviewed and labelled by myself with considerable findings, where Signal Harbour's buckets match my labels on 60.8% of the full sample (56.0% development, 68.9% holdout; n=120). The automated pseudo-label column stays a diagnostic only, and these are polarity-match rates, not precision or recall. Since 61.7% of my labels are positive, an always-positive rule would also score about 61.7%, so the holdout figure is the more informative one. Figure A9 instead isolates the components that move scores. On the 800-headline ablation sample, *no_neutral_overrides* matches *full_signal_harbour*, suggesting that the retained masks rarely activate, while added terms and phrases do change the neutral share. Figures A10 and A11 keep active-ticker counts and coverage separate from model behaviour. Nonetheless, the coverage is uneven, with Technology averaging about 6.1 headlines per sector-day and Consumer about 5.5, while Materials and Real Estate sit near 2.0 and 1.9. This means a thin sector's

daily reading can rest on one or two headlines plus neutral fill, deserving less trust, which is why Signal Harbour down-weights sparse names through coverage confidence.

Table 4. Model polarity shares on 4,000 headlines. The comparison is distributional only. Source: *sentiment_model_comparison.csv*; Figure A8.

Model	Mean	Positive	Neutral	Negative
Base VADER	0.106	37.5%	49.6%	12.9%
Week 9 finVADER	0.069	26.6%	62.1%	11.3%
Signal Harbour	0.081	29.8%	58.7%	11.6%

Fusion result

The main fusion test applies the lagged context-weighted score to Equity Minimum Variance. Figure A6 shows almost overlapping wealth paths, with a small deterioration after the tilt: Sharpe falls from 0.4740 to 0.4711 and CAGR from 5.36% to 5.32%. Changing the tilt strength or the underlying equity method does not reverse the result, and none of the specifications in Figure A7 and Table 6 exceeds Equity Equal Weight. Under the equal turnover charges in Table 5, fusion remains behind and the gap widens slightly. The score still provides a continuous lagged monitoring series, but it does not add alpha in this sample.

Table 5. Base versus fusion under equal turnover costs. TO is average annual turnover. Source: *fusion_cost_sensitivity.csv*.

Sleeve	Cost	Sharpe	CAGR	TO	Terminal
Base	0 bp	0.474	5.36%	1.90	1.169x
Fusion	0 bp	0.471	5.32%	1.91	1.167x
Base	5 bp	0.467	5.26%	1.90	1.165x
Fusion	5 bp	0.464	5.22%	1.91	1.164x
Base	10 bp	0.459	5.16%	1.90	1.162x
Fusion	10 bp	0.456	5.12%	1.91	1.161x
Base	25 bp	0.437	4.86%	1.90	1.152x
Fusion	25 bp	0.434	4.82%	1.91	1.151x

Table 6. Fusion robustness across equity bases and tilt strengths at zero cost. Source: *fusion_sensitivity.csv*; Figure A7.

Base	Tilt	CAGR	Sharpe	MaxDD
EW	0.15	13.24%	0.853	-20.2%
EW	0.35	13.17%	0.849	-20.2%
EW	0.60	13.08%	0.845	-20.3%
RP	0.15	10.39%	0.753	-18.4%
RP	0.35	10.34%	0.751	-18.4%

Base	Tilt	CAGR	Sharpe	MaxDD
RP	0.60	10.29%	0.748	-18.5%
MinVar	0.15	5.34%	0.473	-14.9%
MinVar	0.35	5.32%	0.471	-14.9%
MinVar	0.60	5.29%	0.469	-14.9%

The portfolio result limits the role assigned to sentiment. Table 4 and Figures A8-A9 show that finance-specific language changes score distributions relative to social-media VADER, and the Explain view traces a lagged signal to its terms and confidence weights. The unsuccessful element is the multiplicative allocation rule over 2021-2023. Headline tone may duplicate price information already present in the estimation window, and the required one-session lag removes the most immediate part of any relationship. Equal Weight and Risk Parity move in the same direction, so this finding is not peculiar to Minimum Variance. Sentiment is retained for monitoring and explanation, not as an independent return engine.

4. The Signal Harbour application

The application is organised around the work an analyst does before an investment-committee meeting. Each tab answers a practical question from retained output files rather than recalculating the research. This keeps the deployment suitable for a free Streamlit service and ensures that a chart shown to a client is drawn from the same evidence reported here.

- **Compare.** Rank sleeves by zero-risk-free-rate Sharpe and terminal growth, then plot a shortlist of wealth paths from the research sample through December 2023.
- **Fact sheet.** Inspect one fund at a time: CAGR, volatility, Sharpe, drawdown, and the latest decision weights. A residual is stated whenever only the top holdings are displayed.
- **Allocate.** Combine completed fund return streams without re-optimising names, using intersecting equity sessions for equity-only blends and the native union calendar for blends containing crypto.
- **Sentiment and Explain.** Review the sector news climate, model comparisons, coverage diagnostics, and the prior-session fields that produced the lagged tradable score. Headlines are not re-scored live.

Calendar treatment in Allocate

Allocate works with completed fund-return series rather than re-optimising the assets inside a Combined fund. When a Crypto fund is selected, the tab uses the union of dates, annualises over 365 days and records zero for the equity leg while that market is closed. Equity-only blends remain on intersecting sessions with 252-day annualisation. A 60/40 equity-crypto mix therefore retains crypto returns earned on Saturday and Sunday instead of deleting them during alignment.

Several interface choices address predictable misreadings. A fixed light theme keeps headings visible in a dark-mode browser. Explain displays the dates attached to the raw sentiment input and the lagged tradable version; the fact sheet reports a residual whenever its top-20 table omits part of the portfolio. These choices leave performance unchanged, but they prevent the display from implying more completeness or timeliness than the underlying output contains.

The client discussion begins with the main investment trade-off. Combined Risk Parity records a 0.91 Sharpe ratio and -19.3% drawdown; Combined Maximum Sharpe adds some growth at the cost of a deeper loss. Only after that comparison does the news view provide context, consistent with sentiment's role as monitoring rather than allocation. If a client questions Energy's strong 2022 reading, the coverage diagnostics are available beside it. Every display is labelled as out-of-sample research through December 2023, not as live production.

5. Recommendations

Three recommendations follow from the fund and fusion results.

- **Default mixed sleeve.** Use Combined Risk Parity as the primary balanced multi-asset comparison within this sample, with a zero-risk-free-rate Sharpe ratio of 0.91 and a maximum drawdown of -19.3% (Table 1; Figures A1, A2 and A4).
- **Satellite growth sleeve.** Keep Combined Maximum Sharpe as an optional higher-growth alternative rather than the default. Terminal wealth reaches 1.55x, but maximum drawdown deepens to -26.2% and the latest holdings are more concentrated (Tables 1 and 3; Figure A3).
- **Sentiment as monitoring, not alpha.** Retain the lagged context-weighted layer in the explainability workflow, but do not market it as a return engine. Relative to Equity MinVar, Sharpe moves from 0.4740 to 0.4711 and deteriorates further under equal cost assumptions (Figure A6; Tables 5 and 6).

6. Critical reflection and limits

The reported rankings follow the stated timing rules, although they remain conditional on a short period and on choices made during model development. They describe this test rather than a permanent ordering of the funds.

Short live window. Equity and Combined performance begins on 5 January 2021 and ends in December 2023. It captures the COVID recovery, the inflation and rate shock of 2022 and the 2023 equity rebound, but not a broad range of market regimes. A longer test could reverse the ordering of Equal Weight and Risk Parity.

Headline results exclude costs. Table 1 compares funds before transaction costs. Table 5 shows that charging 5-25 basis points per unit of turnover lowers fusion performance, yet the same grid has not been applied to every fund. It therefore remains unclear how much of Maximum Sharpe's higher growth would survive against a lower-turnover Risk Parity strategy.

Uneven news coverage. Materials and Real Estate contain fewer headlines and more exact-zero raw scores than Technology or Consumer (Figure A11). A movement in either sparse sector consequently rests on less information than a similar movement in a densely covered sector.

Distributional, sentiment evidence. Polarity shares and the Week 9 ablation identify how the scores change, but they do not measure classification accuracy.

End-point holdings are not path explanations. Tables 2 and 3 describe the final December 2023 decision. They are useful evidence of end-point concentration, but they cannot explain returns earned earlier in the 2021-2023 path.

Product-level reflection

Using the actual Week 9 analyser, rather than attaching a small dictionary to plain VADER, keeps the extension connected to the course baseline. It also clarifies provenance: course lexicons remain separate from the project additions, and the application openly reads precomputed files. The interface therefore reflects what the system actually does instead of appearing to run a live model.

The labelled evidence has also been a small worksheet, which means it is not a precision study with only 120 labels. This means the agreement figures are polarity-match rates on a considerable small sample and therefore support no precisions or recall claims. Operationally, I would test whether the lagged sector index actually lines up with the subsequent sector returns, not just stopping at the new climate description in A5. These steps do not alter the present finding: the tested sentiment overlay is economically flat to negative in this sample.

What out-of-sample means here

Walk-forward weights use past returns, and tradable sentiment is delayed by one equity session. Those rules protect the timing of individual portfolio returns, but they do not turn the whole research process into a sealed experiment. Lexicon retention, tilt strength and the base portfolio were considered with knowledge of the study period. The negative fusion result therefore applies to this overlay design; it cannot establish that every news rule would fail on an untouched sample. The development-holdout split will support later headline labelling, but it is not yet a pristine test set.

Signal Harbour joins the systematic fund menu to the news features developed in Part A. For a small advisory desk, the mixed-asset evidence favours Combined Risk Parity; sentiment is useful for context and monitoring, not as a source of alpha. Calendar, cost and coverage limitations remain visible because each can change the decision. The final document version is also deployed, and reproduces the submitted output.

Appendix A. Exhibits

Each exhibit is discussed in Sections 2 to 4. Captions restate the sample, source, and the claim the figure is meant to support.

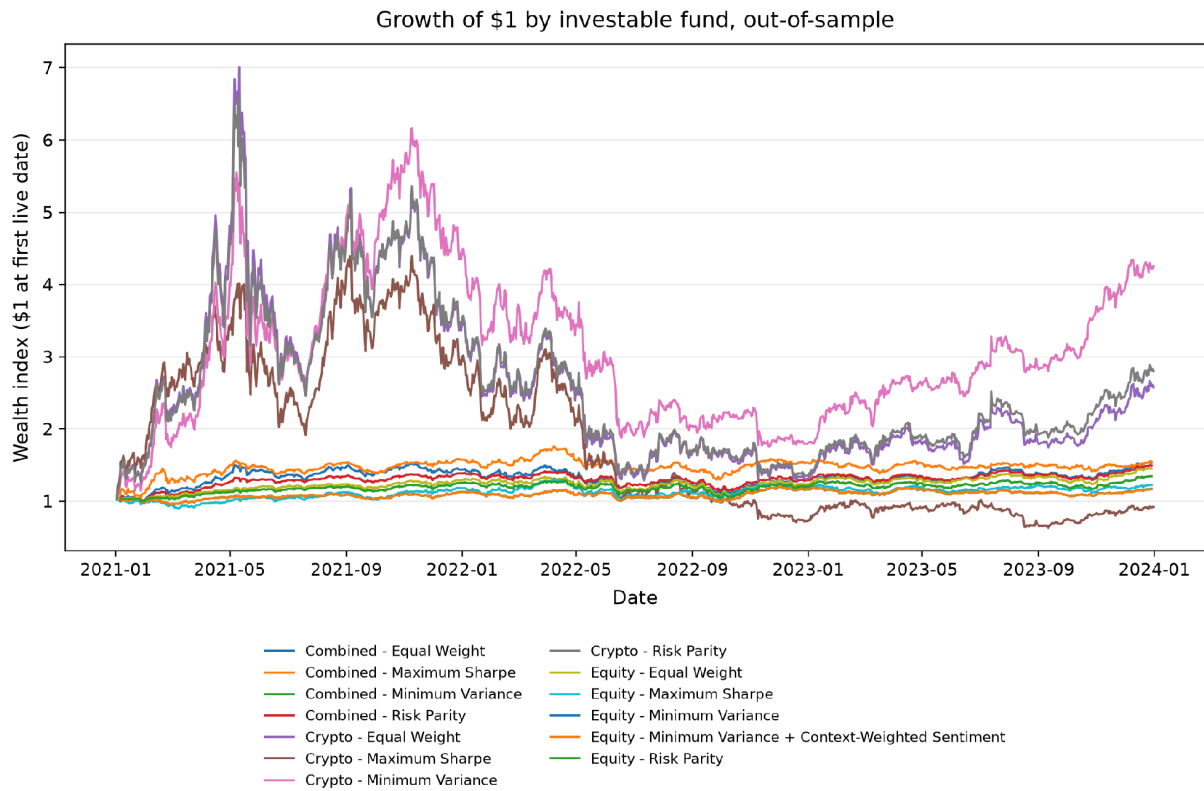


Figure A1. Growth of \$1 across funds, out-of-sample through December 2023. Combined Risk Parity is steadier than Combined Maximum Sharpe; Crypto Minimum Variance leads terminal wealth among the crypto sleeves. Source: fund_returns.csv.

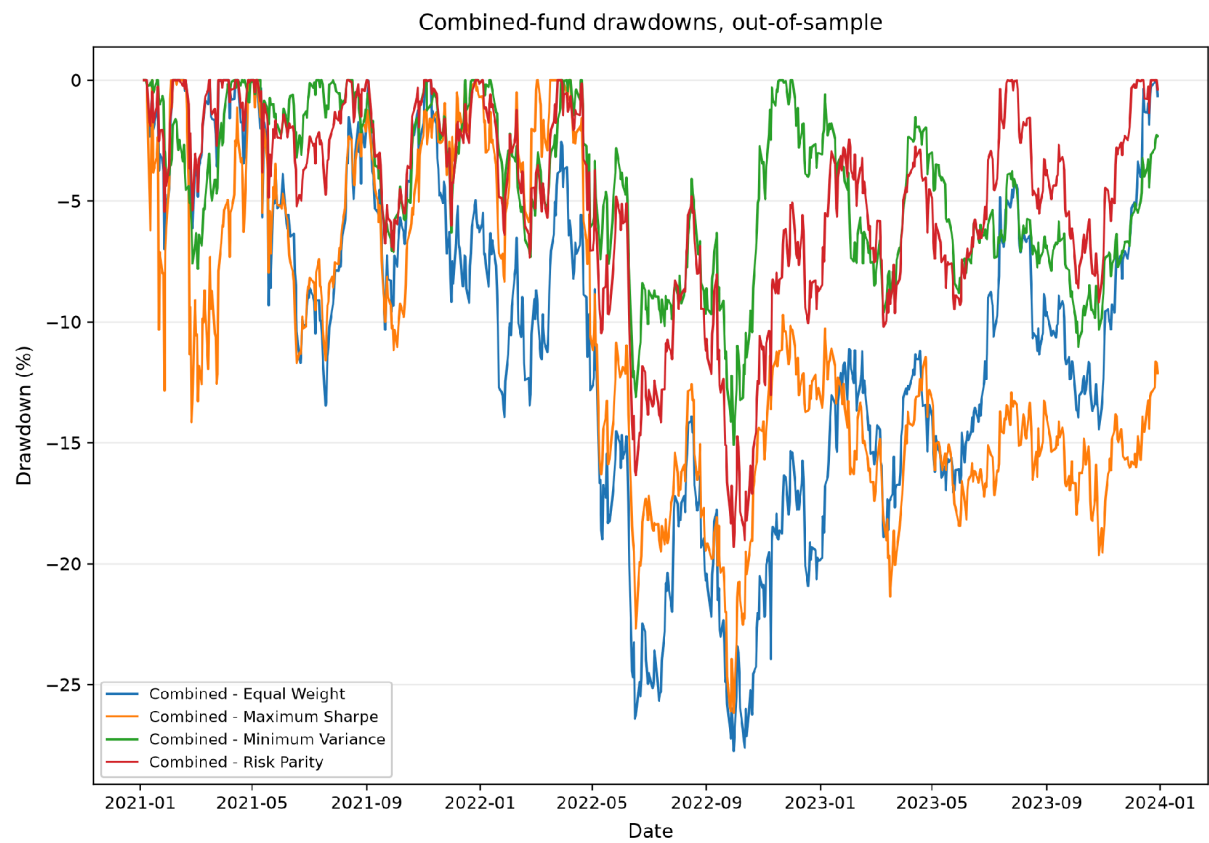


Figure A2. Combined-fund drawdowns. Risk Parity has the shallower path, supporting Recommendation 1; Maximum Sharpe reaches the deeper trough expected of the higher-growth satellite. Wealth paths prepend 1.0 so that the first loss is visible.

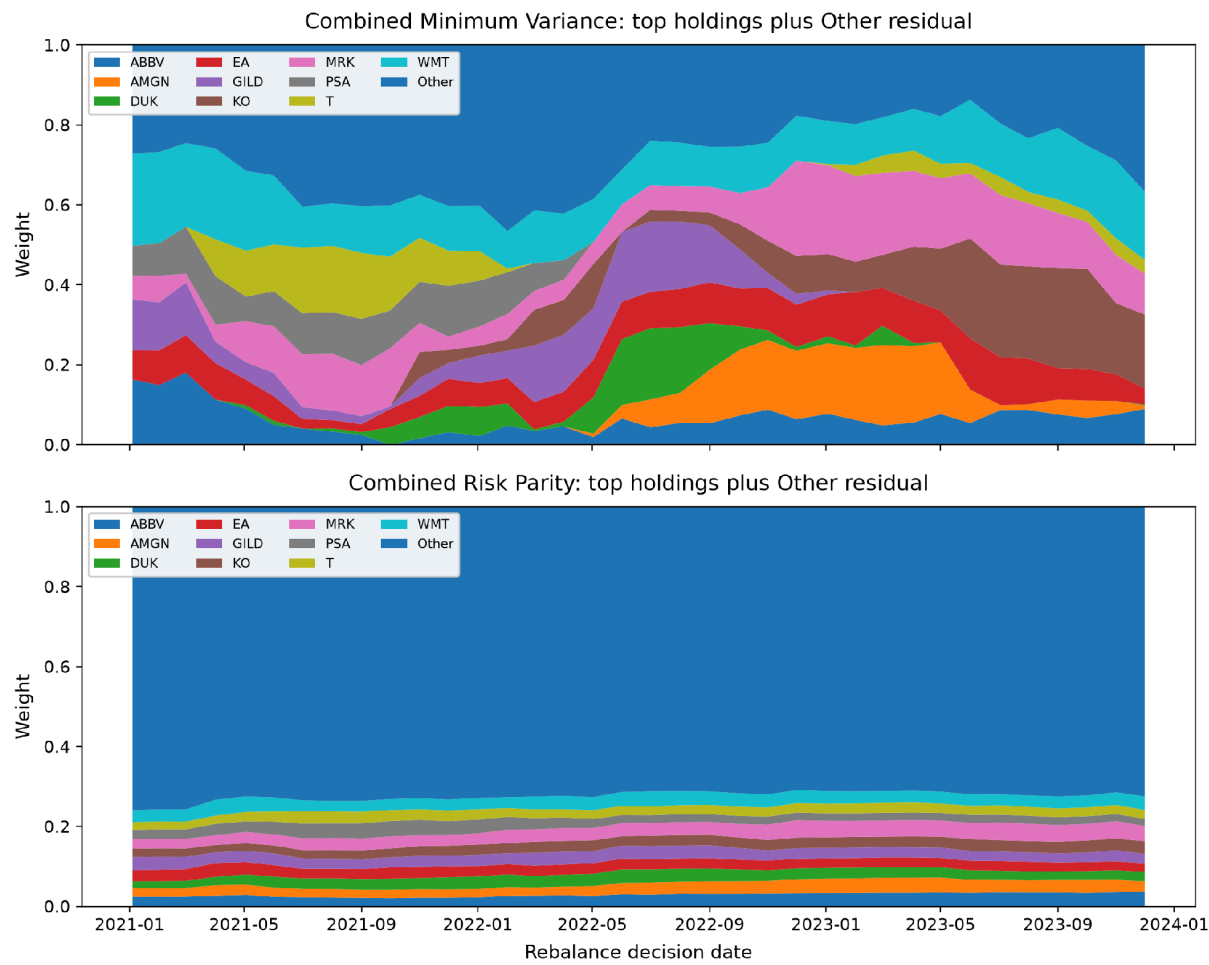


Figure A3. Combined Minimum Variance and Risk Parity decision weights, with an Other residual so the stacked areas sum to one. Maximum Sharpe concentration is summarised in Table 3 rather than in this panel.

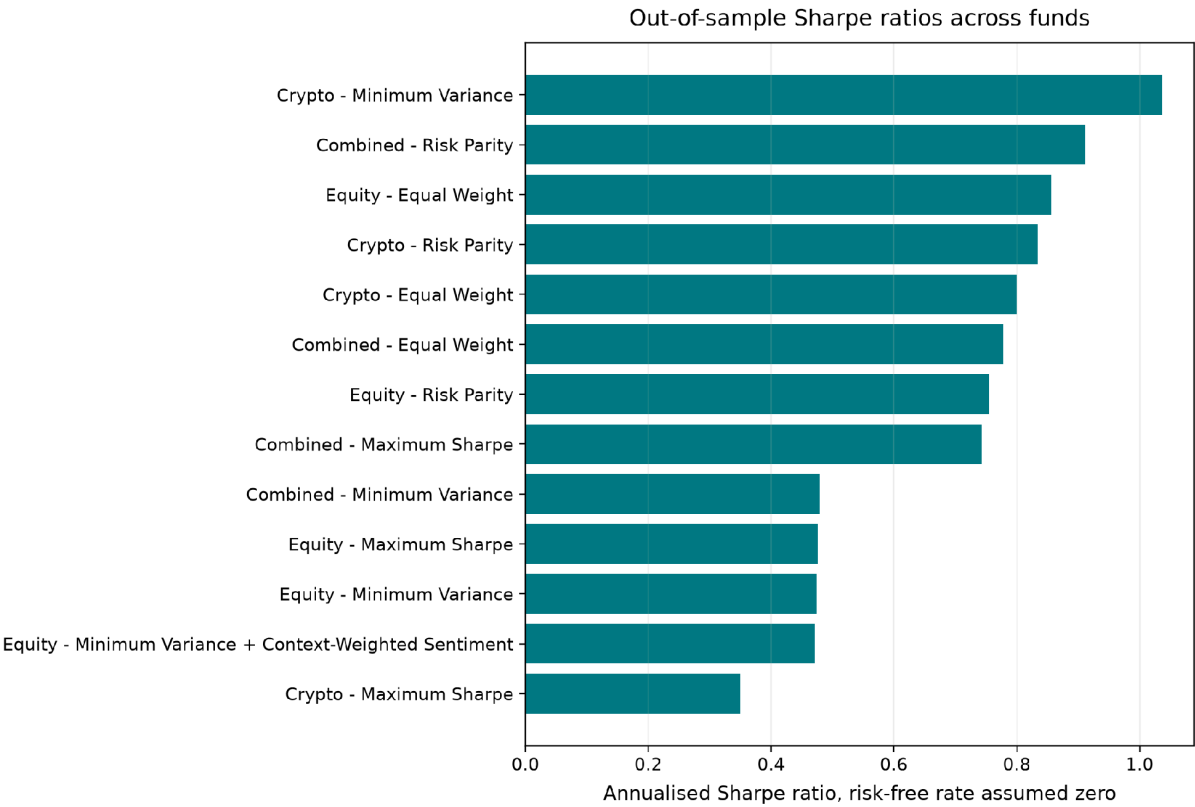


Figure A4. Zero-risk-free-rate Sharpe rankings. Combined Risk Parity leads the mixed sleeves; Equity Equal Weight leads equities; Crypto Minimum Variance leads crypto. Annualisation is 252 or 365 days by sleeve.

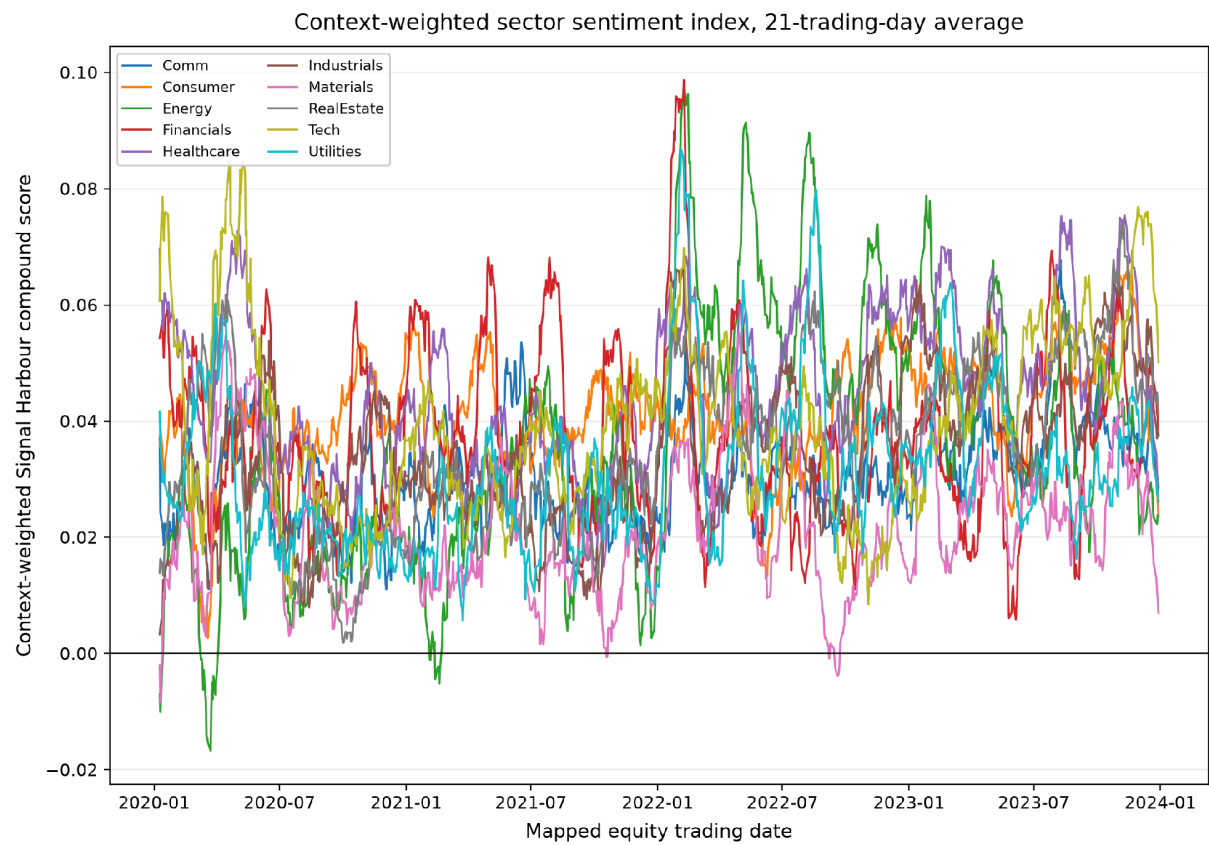


Figure A5. Context-weighted sector sentiment, shown as a 21-session average. The series uses the complete ticker-day grid, sets no-news observations to neutral, and is lagged before any trading use.

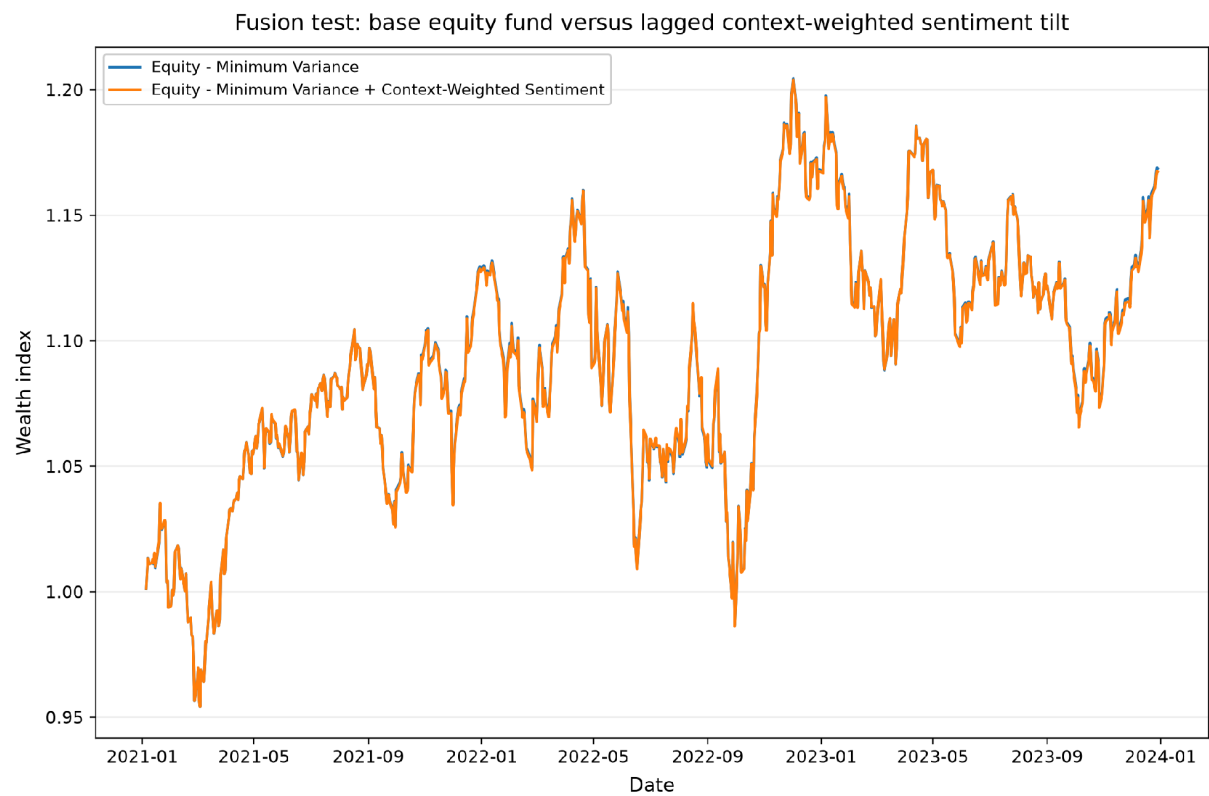


Figure A6. Equity Minimum Variance against the lagged context-weighted sentiment tilt. The overlay does not improve the corrected baseline, which supports Recommendation 3.

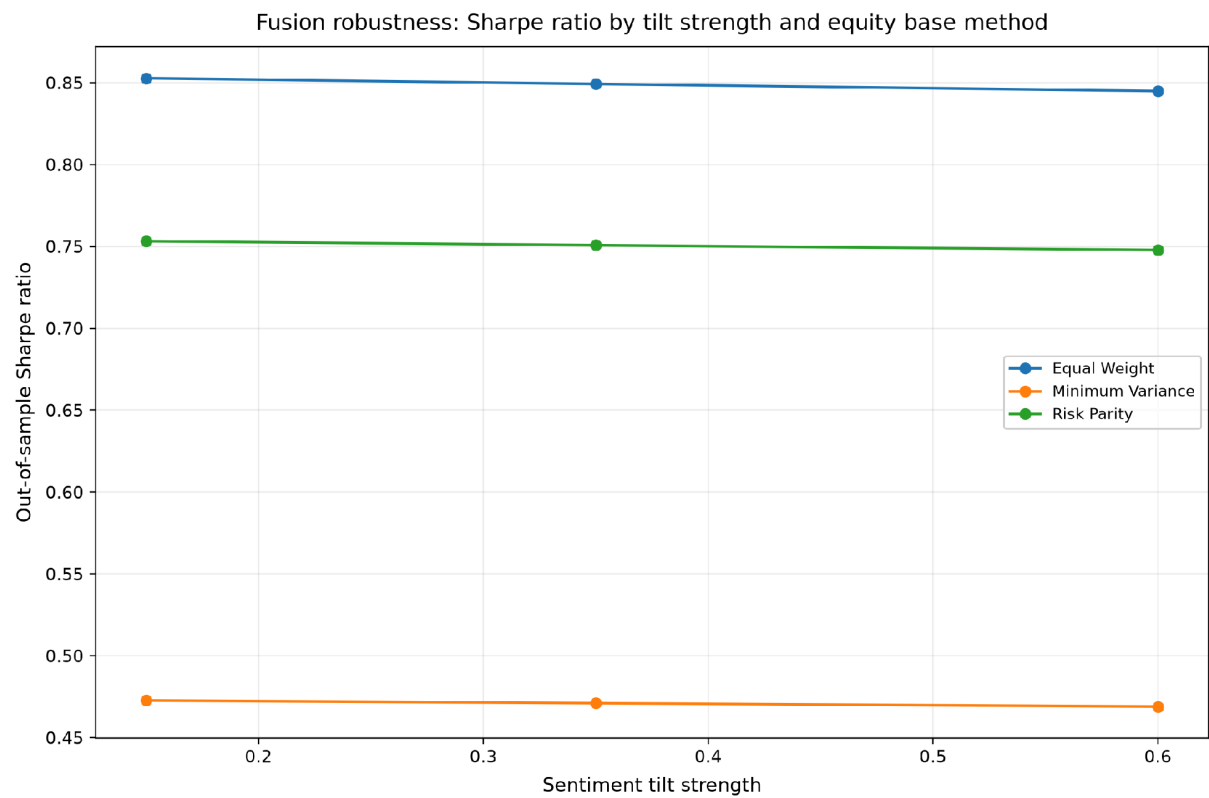


Figure A7. Fusion sensitivity by tilt strength and equity base at zero cost. None of the tested specifications overtakes Equity Equal Weight (Table 6).

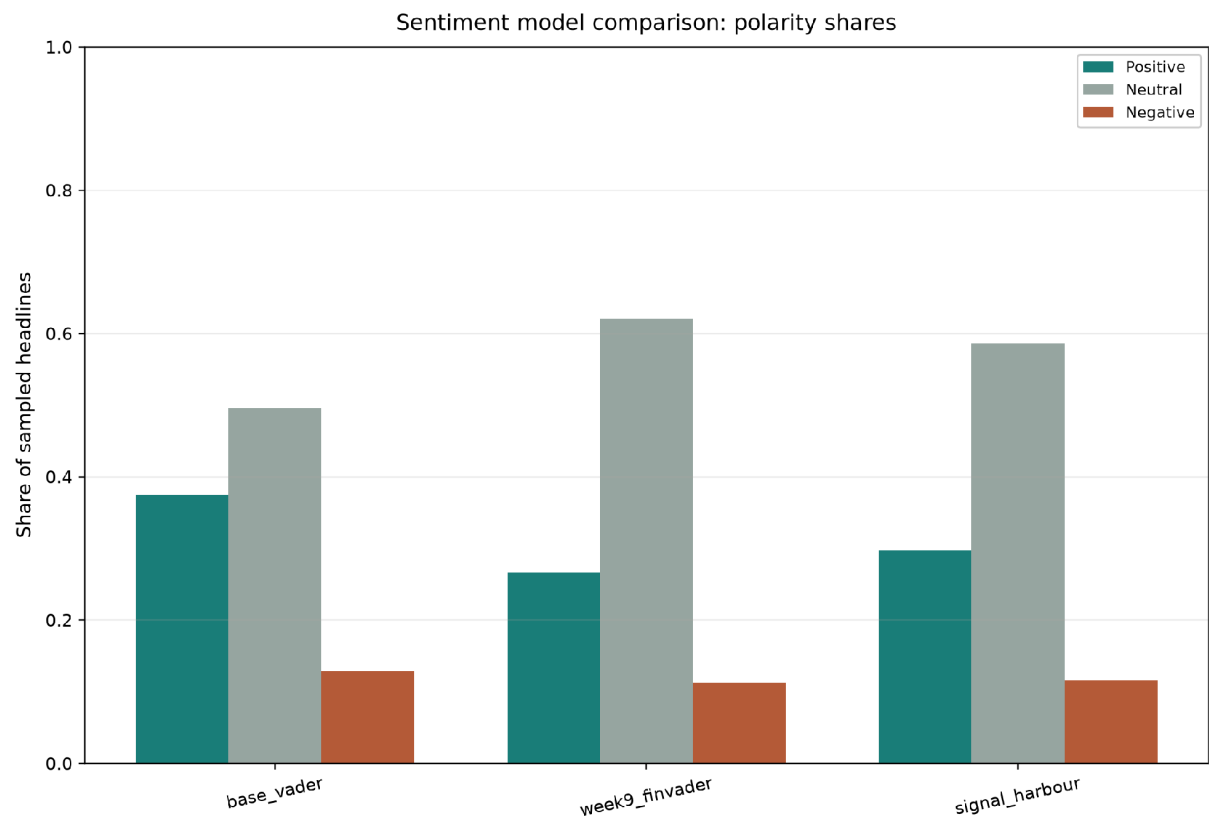


Figure A8. Polarity shares for Base VADER, Week 9 finVADER, and Signal Harbour on 4,000 headlines (Table 4). The comparison is distributional only.

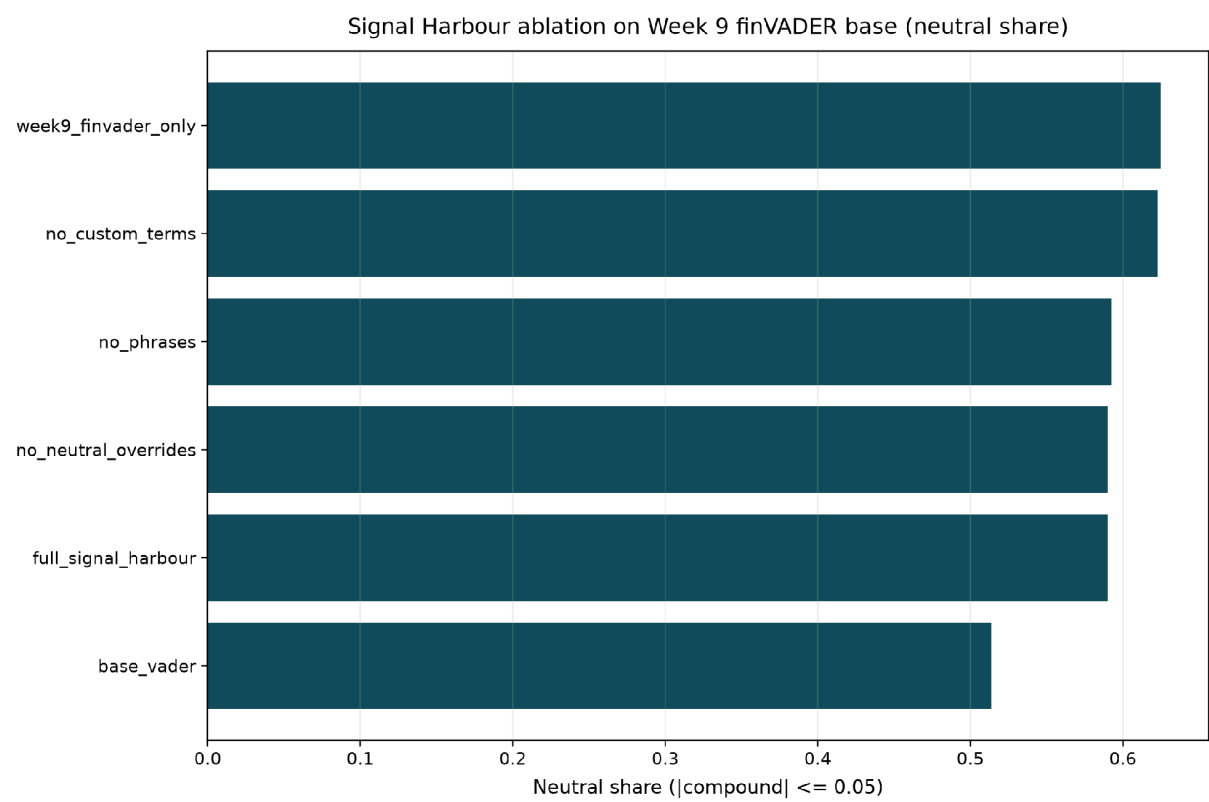


Figure A9. Ablation of Signal Harbour components on the Week 9 finVADER base analyser, with Week 9-only and plain-VADER references.

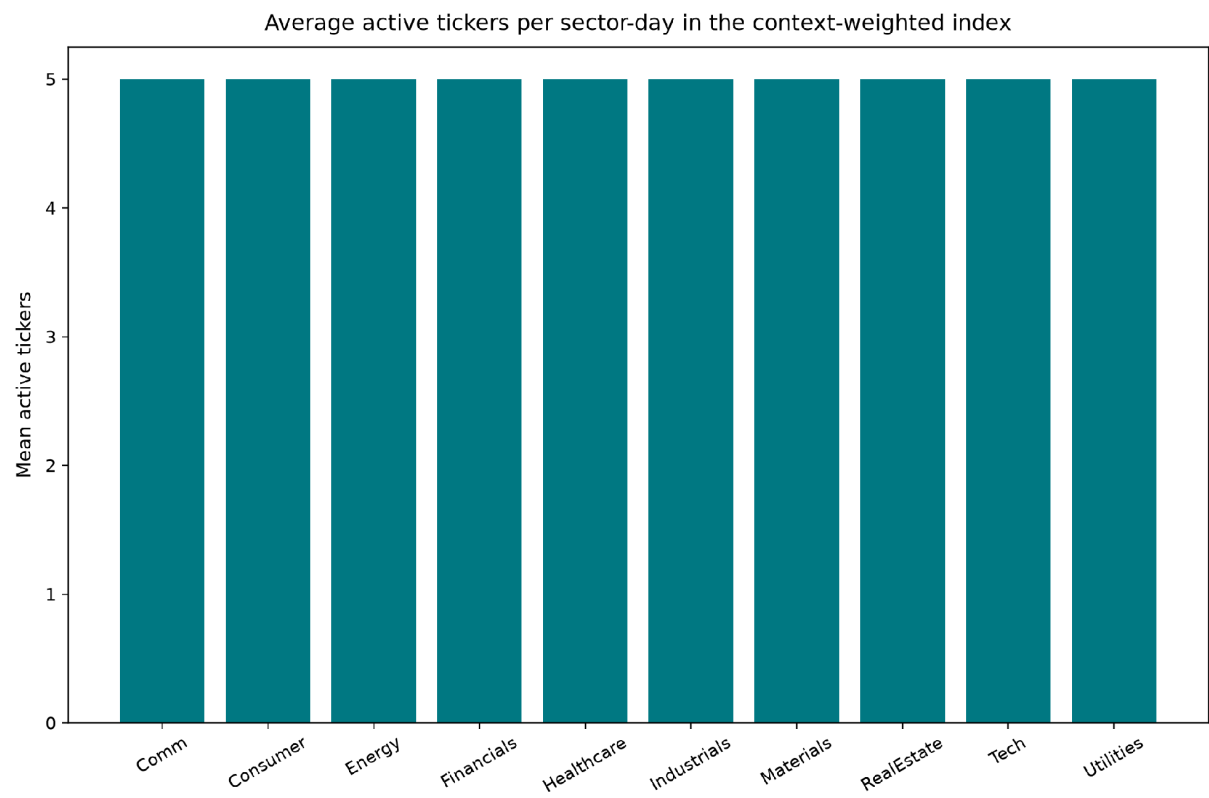


Figure A10. Average active tickers per sector-day under the complete-grid context-weighted index. This is the coverage companion to Figure A5.

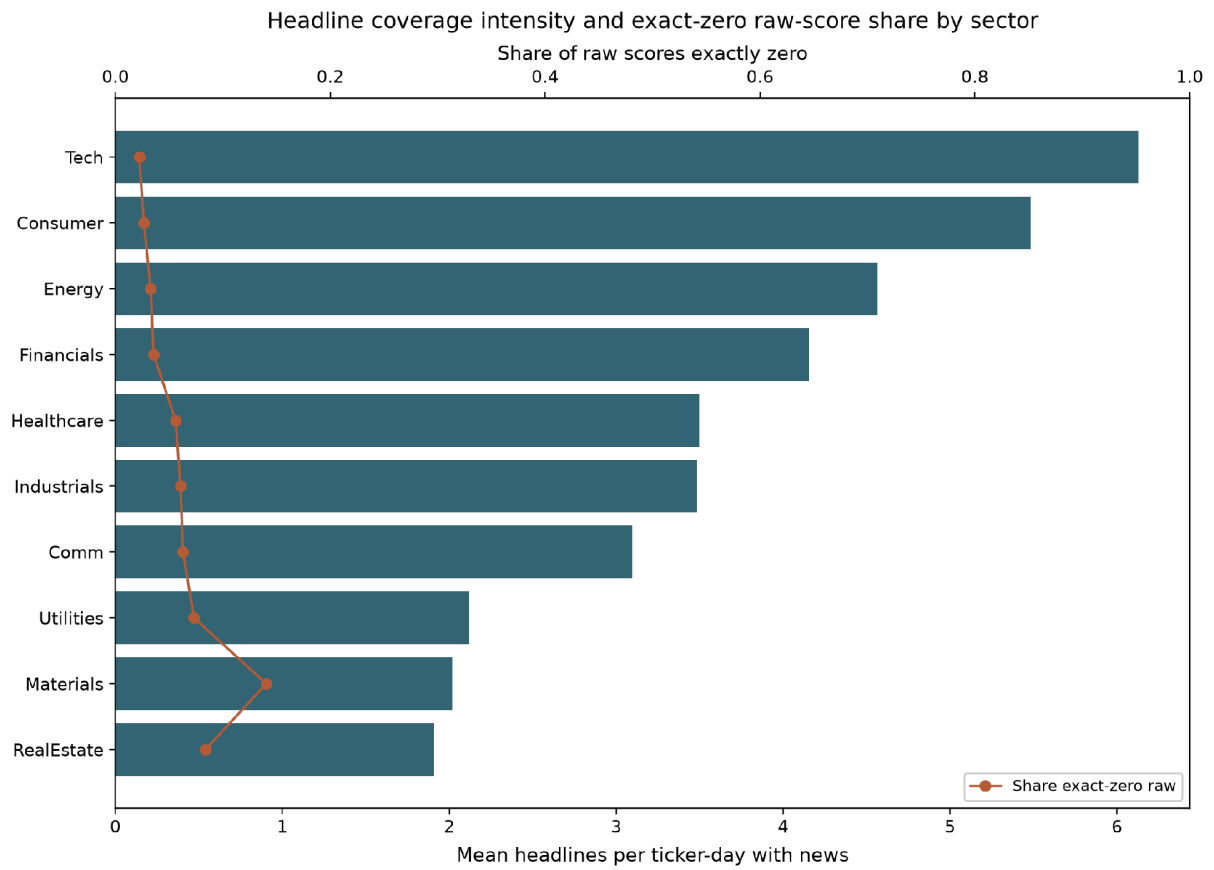


Figure A11. Sector coverage intensity: mean headlines on news-bearing ticker-days (bars) and the share of exact-zero raw scores (line). This exhibit is distinct from the polarity comparison in Figure A8.

Appendix B. Core equations and references

Minimum variance

minimise $w^T \Sigma_{\text{Ann}} w$ subject to $1^T w = 1$ and $0 \leq w(i) \leq w_{\text{Max}}$

$\Sigma_{\text{Ann}} = \text{ppy} * \text{Cov}(R)$, where ppy is 252 on the equity calendar or 365 on the native crypto calendar.

Maximum Sharpe

maximise $(w^T \mu_{\text{Ann}}) / \sqrt{w^T \Sigma_{\text{Ann}} w}$

The risk-free rate is zero; the long-only and name-cap constraints are unchanged.

Reported performance

$\text{Sharpe} = (\text{mean}(r) * \text{ppy}) / (\text{std}(r) * \sqrt{\text{ppy}})$

$\text{CAGR} = \text{WT}$ raised to the power (ppy/T) , minus 1

Context-weighted score

$sCw(i, t) = s(i, t) * \text{coverageConf}(i, t) * \text{attentionConf}(i, t)$

Tradable score = $sCw(i, t)$ minus 1).

Sentiment tilt

w_{Tilt} is proportional to $w * \exp(\lambda * (\text{score} - \text{mean}(\text{score})))$. The tilted vector is then projected back onto the capped simplex.

References

- Hutto, C. J., and Gilbert, E. (2014). VADER: A parsimonious rule-based model for sentiment analysis of social media text. *ICWSM*.
- Korab, P. FinVADER lexicons (SentiBigNomics / Henry path used in FINS3645 Week 9). Apache 2.0; vendored under `src/vendor/course_finvader`.
- Loughran, T., and McDonald, B. (2011). When is a liability not a liability? *The Journal of Finance*, 66(1), 35 to 65.
- Markowitz, H. (1952). Portfolio selection. *The Journal of Finance*, 7(1), 77 to 91.
- FINS3645 Project 2026 brief and data access helper; Kaiyuan Lan, Project A Signal Harbour / Attention Pulse (2026).